



正基科技股份有限公司

SPECIFICATION

PRODUCT NAME : AP6236

REVISION : 2.1

DATE : Nov 19th, 2019

Customer APPROVED	
Company	
Representative Signature	

PREPARED	REVIEW			APPROVED	DCC ISSUE
	PM	QA	ET		



正基科技股份有限公司



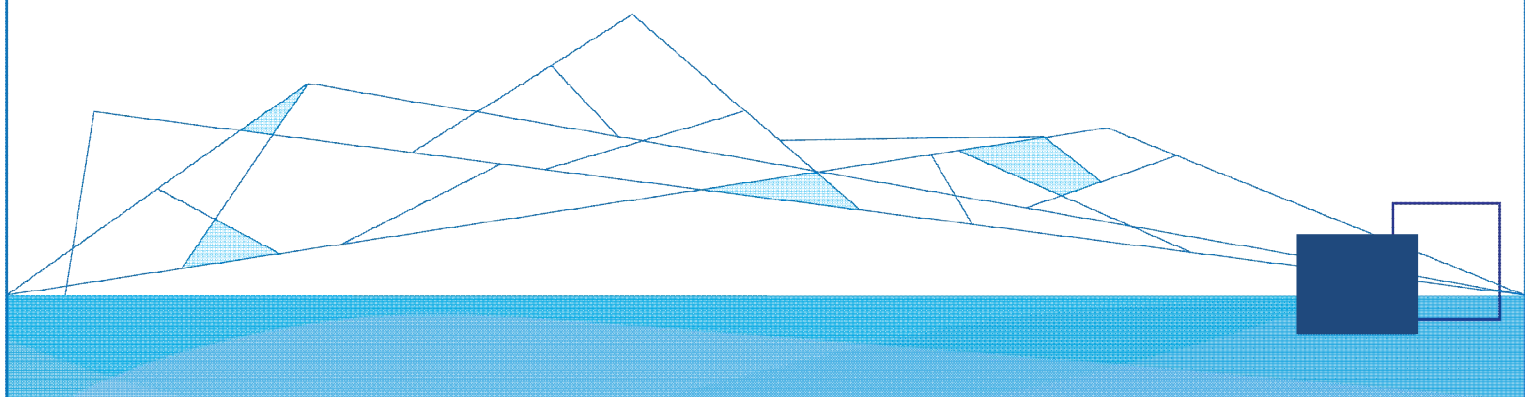
AP6236 Data Sheet

Address:

3F., No.15-1 Zhonghua Rd, Hsinchu Industrial Park, Hukou, Hsinchu,
Taiwan, 30352

Website:

<http://www.ampak.com.tw>



Revision

Revision	Date	Description	Revised By
1.0	2017/06/15	- Preliminary	Richard
1.1	2017/09/01	- Modify Recommended Reflow Profile	Richard
1.2	2018/02/07	- Modify 2.4GHz support channel	Richard
1.3	2018/04/10	- Modify Pin Assignments	Richard
1.4	2018/07/19	- Modify GPIO	Ali
1.5	2018/08/20	- Modify SDIO/gSPI strapping. - Modify gSPI Mode timing	Richard
1.6	2018/09/10	- Modify External LPO signal characteristics	Richard
1.7	2018/09/13	- Modify operating temperature range	Ali
1.8	2018/12/10	- Modify 4.2 Voltages - Modify PCM timing Diagram	Ali
1.9	2019/07/03	- Modify Bluetooth Specification	Richard
2.0	2019/09/20	-Format Update	Richard
2.1	2019/11/19	-Modify Module Dimensions	Ali

Contents

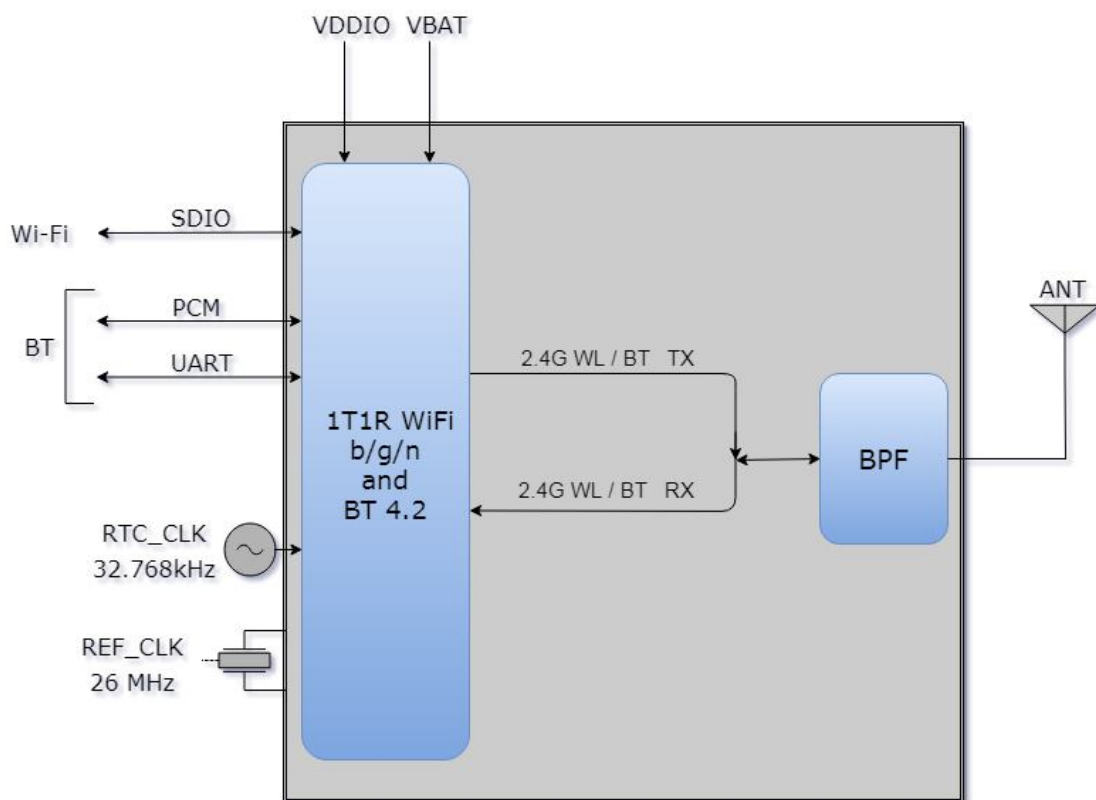
DCC ISSUE	0
Revision	1
1. Introduction	2
1.1 Overview	2
1.2 Product Features	3
2. General Specification	4
2.1 General Specification	4
2.2 DC Characteristics	4
2.2.1 Absolute Maximum Ratings	4
2.2.2 Recommended Operating Rating	5
The module requires two power supplies: VBAT and VDDIO.	5
The module requires two power supplies: other Digital I/O Pins.	5
3. Wi-Fi RF Specification	6
3.1 2.4GHz RF Specification	6
4. Bluetooth Specification	8
4.1 Bluetooth Specification	8
5. Pin Definition	9
5.1 Pin Outline	9
5.2 Pin Assignment	10
6. Dimensions	12
6.1 Module Dimensions	12
6.2 Recommended Footprint	13
7. External clock reference	14
8. Host Interface Timing Diagram	15
8.1 Power-up Sequence Timing Diagram	15
8.2 SDIO Interface Description	17
8.3 gSPI Mode Timing Diagram	20
8.5 PCM Interface Description	21
8.6 UART Interface Description	27
9. Recommended Reflow Profile	29
10. Package Information	30
10.1 Label	30
10.2 Dimension	31
10.3 MSL Level / Storage Condition	33

1. Introduction

1.1 Overview

The AMPAK Technology® AP6236 is a fully Wi-Fi and Bluetooth functionalities module with seamless roaming capabilities and advance security, also it could interact with different vendors' 802.11b/g/n 1x1 Access Points with SISO standard and can accomplish up to speed of 72Mbps with single stream in 802.11n to connect the wireless LAN. Furthermore AP6236 included SDIO interface for Wi-Fi, UART/ PCM interface for Bluetooth.

In addition, this compact module is a total solution for a combination of Wi-Fi + BT technologies. The module is specifically developed for tablet, OTT box and portable devices.



1.2 Product Features

wifi:2.4G

- Lead Free design which is compliant with ROHS requirements.
 - Single-stream spatial multiplexing up to 72 Mbps data rate.

72Mbps
 - Supports 1antenna with one for WLAN & Bluetooth shared port. Also, shared Bluetooth and WLAN receive signal path eliminates the need for an external power splitter while maintaining excellent sensitivity for both Bluetooth and WLAN.
 - Supports standard SDIO v2.0
 - BT host digital interface:
 - HCI UART (up to 4 Mbps)
 - PCM for audio data

蓝牙4.2
 - Complies with Bluetooth Core Specification Version 4.2 with provisions for supporting future specifications. With Bluetooth Class 1 or Class2 transmitter operation.
 - Supports extended synchronous connections (eSCO), for enhanced voice quality by allowing for retransmission of dropped packets.
 - Adaptive frequency hopping (AFH) for reducing radio frequency interference.
- A simplified block diagram of the module is depicted in the figure above.

2. General Specification

2.1 General Specification

Model Name	AP6236
Product Description	1Tx/1Rx 802.11 b/g/n Wi-Fi + BT 4.2 Module
Dimension	L x W: 12 x 12(Typ.)mm 、 H : 1.65 (Max.) mm
WiFi Interface	SDIO V 2.0
BT Interface	UART / PCM
Operating temperature ^{a,b}	-30°C to 85°C
Storage temperature	-40°C to 125°C
Humidity	Operating Humidity 10% to 95% Non-Condensing

- a. Functionality is guaranteed across this range of temperature. Optimal RF performance as specified in the data sheet, however, is guaranteed only for -10°C to 55°C.
- b. The operation temperature 65°C to 85°C is feasible at conditional environment. Please examine the reliability on final product.

2.2 DC Characteristics

2.2.1 Absolute Maximum Ratings

Symbol	Description	Min.	Max.	Unit
VBAT	Input supply Voltage	-0.5	5	V
VDDIO	Digital/Bluetooth/SDIO/ I/O Voltage	-0.5	3.9	V

Extreme caution must be exercised to prevent electrostatic discharge (ESD) damage.

Symbol	Condition	Minimum ESD Rating	Unit
ESD_HAND_HBM	Human body model contact discharge per JEDEC EID/JESD22-A114	1	kV
ESD_HAND_CDM	Charged device model contact discharge per JEDEC EIA/JESD22-C101	250	V

2.2.2 Recommended Operating Rating

The module requires two power supplies: VBAT and VDDIO.

Voltage rails	Min.	Typ.	Max.	Unit
VBAT	3.2	3.3	3.8	V
VDDIO	1.6	1.8 / 3.3	3.6	V

VBAT current consumption 1A (Peak), when VBAT = 3.3V

The module requires two power supplies: other Digital I/O Pins.

For VDDIO=1.8V	Min.	Max.	Unit
VIL/VIH	$0.35 \times VDDIO$	$0.65 \times VDDIO$	V
VOL/VOH output@2mA	0.4	VDDIO-0.4	V
For VDDIO=3.3V	Min.	Max.	Unit
VIL/VIH	0.80	2	V
VOL/VOH output@2mA	0.4	VDDIO-0.4	V

3. Wi-Fi RF Specification

3.1 2.4GHz RF Specification

Conditions : VBAT=3.3V ; VDDIO=3.3V ; Temp:25°C

Feature		Description			
WLAN Standard		IEEE 802.11b/g/n & Wi-Fi compliant			
Frequency Range		2.400 GHz ~ 2.4835 GHz (2.4GHz ISM Band)			
Number of Channels		2.4GHz：Ch1 ~ Ch13			
Modulation		802.11b：DQPSK、DBPSK、CCK 802.11 g/n：OFDM /64-QAM、16-QAM、QPSK、BPSK			
Output Power , tolerance ± 1.5 dB					
The transmit EVM quality & spectrum mask are compliant with IEEE 802.11 standard					
802.11b	1Mbps	2Mbps	5.5Mbps	11Mbps	
	17	17	17	17	
802.11g	6、9Mbps	12、18Mbps	24Mbps	36Mbps	48Mbps
	17	17	16	16	15
	54Mbps				
	15				
802.11n 20MHz	MCS0~2	MCS3	MCS4	MCS5	MCS6
	17	17	16	15	14
	MCS7				
	14				
Note：The specifications of RF output power are subject to change to fulfill the safety regulation and requirements in end-user product.					
Sensitivity, tolerance ± 2 dB					
CCK modulation PER ≤ 8%、OFDM modulation PER ≤ 10%					
802.11b	Data Rate	Spec.(dBm)			
	1Mbps	-95			
	2Mbps	-92			
	5.5Mbps	-90			
	11Mbps	-88			
802.11g	Data Rate	Spec.(dBm)	Data Rate	Spec.(dBm)	
	6Mbps	-89	24Mbps	-82	
	9Mbps	-87	36Mbps	-79	
	12Mbps	-86	48Mbps	-76	

	18Mbps	-84	54Mbps	-73
802.11n_20MHz	Data Rate	Spec.(dBm)	Data Rate	Spec.(dBm)
	MCS0	-88	MCS4	-79
	MCS1	-86	MCS4	-75
	MCS2	-84	MCS6	-73
	MCS3	-82	MCS7	-71
Maximum Input Level	802.11b : -10 dBm			
	802.11g/n : -20 dBm			

AMPAK
Confidential

4. Bluetooth Specification

4.1 Bluetooth Specification

Conditions : VBAT=3.3V ; VDDIO=3.3V ; Temp:25°C

Feature	Description	
General Specification		
Bluetooth Standard	BDR 、EDR(1Mbps & 2Mbps) 、LE(1Mbps)	
Host Interface	UART	
Frequency Band	2402 MHz ~ 2480 MHz	
Number of Channels	79 channels for classic 、40 channels for BLE	
Modulation	GFSK, DPSK, DQPSK	
RF Specification		
	CL1 (dBm)	CL2 (dBm)
BDR Output Power	6	2
EDR Output Power	4	2
LE Output Power	5	2
	Typical.	
Sensitivity @ BER=0.1% for GFSK (1Mbps)	-86 dBm	
Sensitivity @ BER=0.01% for $\pi/4$ -DQPSK (2Mbps)	-87 dBm	
Sensitivity @ BER=0.01% for 8DPSK (3Mbps)	-83 dBm	
Sensitivity @ BER=30.8% for LE (1Mbps)	-88 dBm	
Maximum Input Level	GFSK (1Mbps):-20dBm	
	$\pi/4$ -DQPSK (2Mbps) :-20dBm	
	8DPSK (3Mbps) :-20dBm	

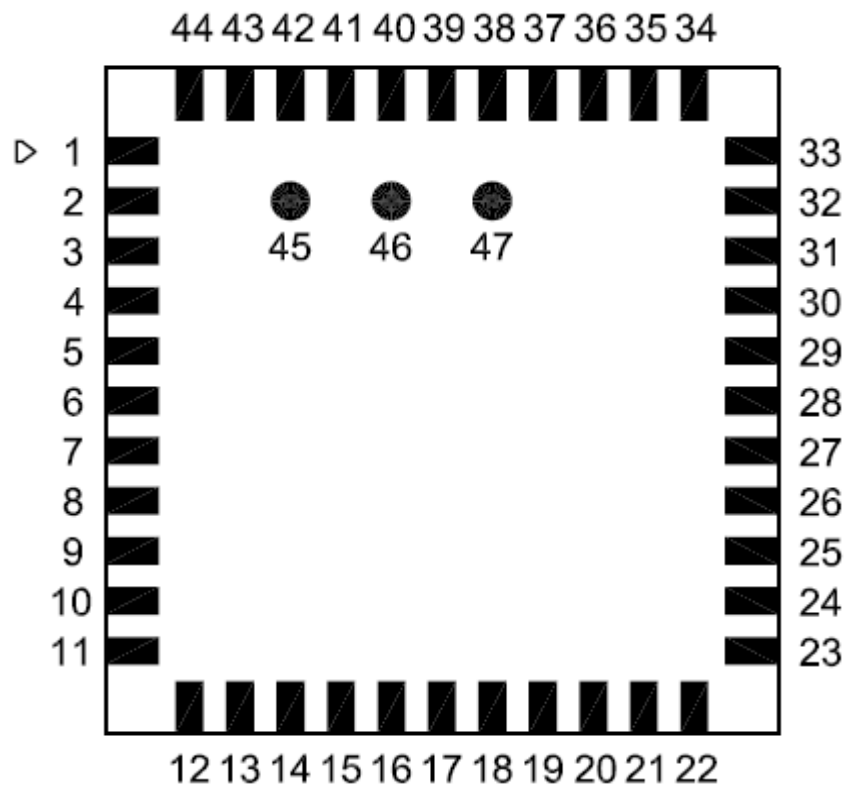
Note* : The Bluetooth output power is able to be configured by firmware (hcd file).



5. Pin Definition

5.1 Pin Outline

<TOP VIEW>



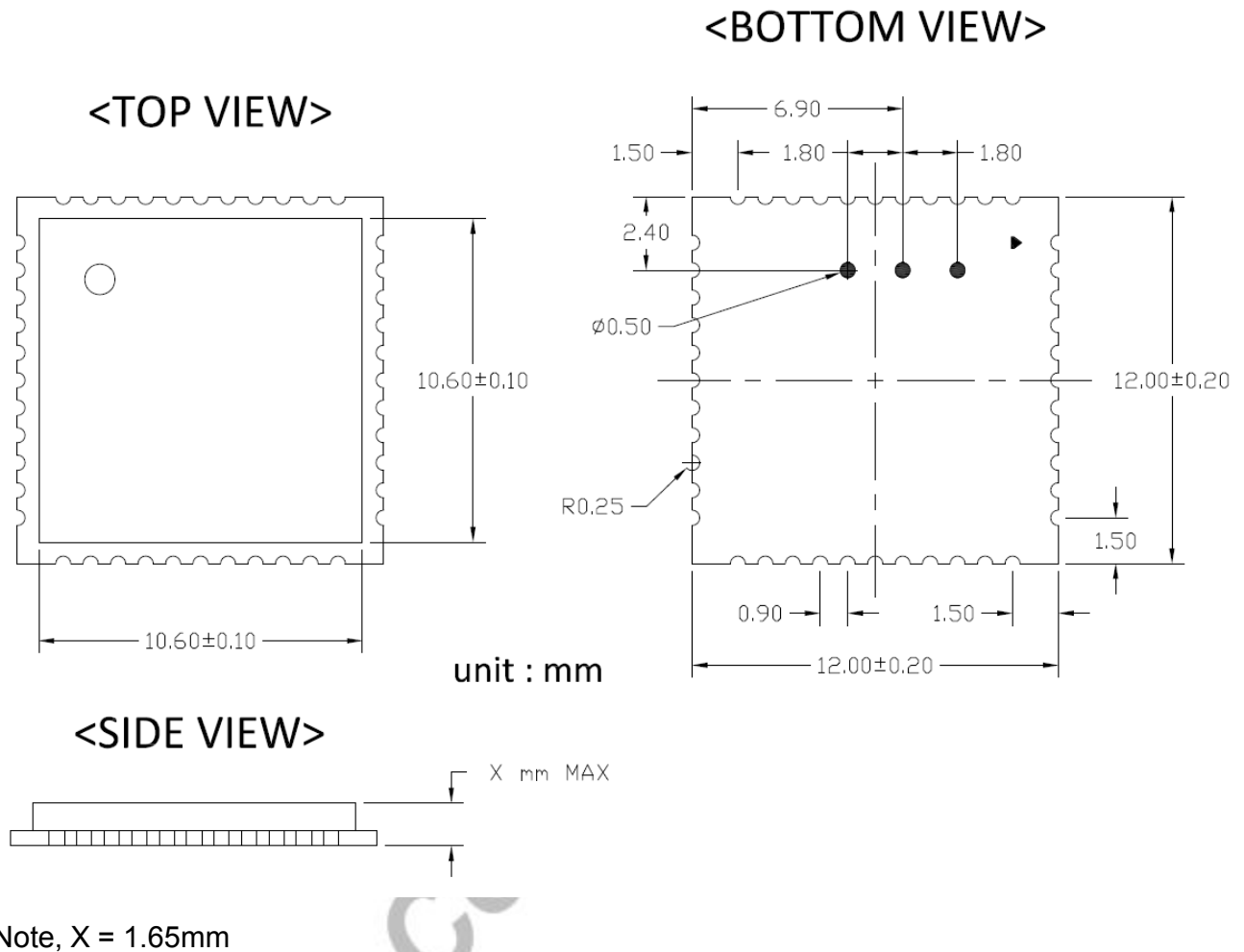
5.2 Pin Assignment

NO	Name	Type	Description
1	GND	—	Ground connections
2	WL_BT_ANT	I/O	RF I/O port
3	GND	—	Ground connections
4	NC	—	Floating (Don't connected to ground)
5	NC	—	Floating (Don't connected to ground)
6	BT_WAKE	I	HOST wake-up Bluetooth device
7	BT_HOST_WAKE	O	Bluetooth device to wake-up HOST
8	NC	—	Floating (Don't connected to ground)
9	VBAT	P	Main power voltage source input
10	XTAL_IN	I	Crystal input
11	XTAL_OUT	O	Crystal output
12	WL_REG_ON	I	Internal regulators power enable/disable
13	WL_HOST_WAKE	O	WLAN to wake-up HOST
14	SDIO_DATA_2/gSPI Strapping	I/O	SDIO data line 2/gSPI Strapping The default is SDIO. For gSPI, pull this pin low
15	SDIO_DATA_3/CS	I/O	SDIO data line 3/ gSPI card select
16	SDIO_DATA_CMD/DI	I/O	SDIO command line/ gSPI data input
17	SDIO_DATA_CLK/SCLK	I/O	SDIO clock line/ gSPI Clock
18	SDIO_DATA_0/DO	I/O	SDIO data line 0/ gSPI data output
19	SDIO_DATA_1/IRQ	I/O	SDIO data line 1/ gSPI interrupt
20	GND	—	Ground connections
21	VIN_LDO_OUT	P	Internal Buck voltage generation pin
22	VDDIO	P	I/O Voltage supply input
23	VIN_LDO	P	Internal Buck voltage generation pin
24	LPO	I	External Low Power Clock input (32.768KHz)
25	PCM_OUT	O	PCM Data output
26	PCM_CLK	I/O	PCM clock
27	PCM_IN	I	PCM data input
28	PCM_SYNC	I/O	PCM sync signal
29	NC	—	Floating (Don't connected to ground)
30	NC	—	Floating (Don't connected to ground)
31	GND	—	Ground connections

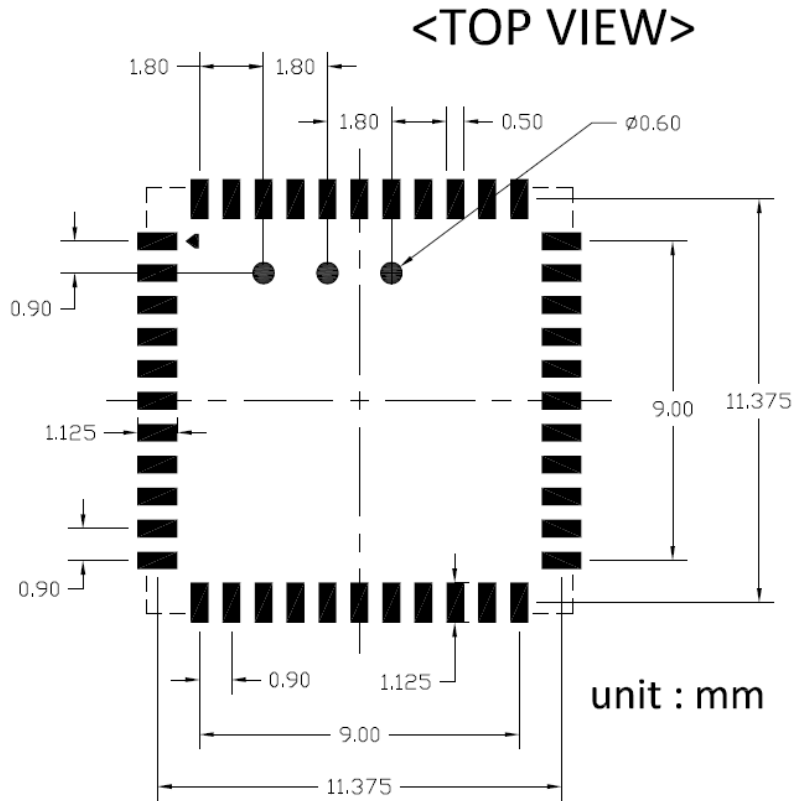
32	NC	—	Floating (Don't connected to ground)
33	GND	—	Ground connections
34	BT_RST_N	I	Low asserting reset for Bluetooth core
35	NC	—	Floating (Don't connected to ground)
36	GND	—	Ground connections
37	NC	—	Floating (Don't connected to ground)
38	NC	—	Floating (Don't connected to ground)
39	GPIO2	I/O	GPIO Control
40	GPIO1	I/O	GPIO Control
41	UART_RTS_N	O	Bluetooth UART interface
42	UART_TXD	O	Bluetooth UART interface
43	UART_RXD	I	Bluetooth UART interface
44	UART_CTS_N	I	Bluetooth UART interface
45	TP1 (NC)	—	Floating (Don't connected to ground)
46	TP2 (NC)	—	Floating (Don't connected to ground)
47	TP3 (NC)	—	Floating (Don't connected to ground)

6. Dimensions

6.1 Module Dimensions

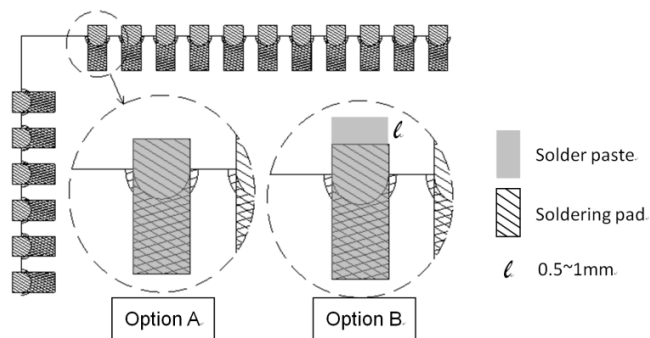


6.2 Recommended Footprint



- Solder paste layer design is generally the same as recommended footprint.
(錫膏層設計通常建議和焊墊尺寸相同)
- If soldering quality with good wetting on upright side is essential for PQC, how to optimize the aperture design in the stencil to adjust the amount of solder paste would be crucial.
In addition, a kind of stencil design with stepped thickness in partial area would be considered if the thickness of stencil is about 0.1mm or thinner. Please optimize the stencil design by manufacture engineer or contact AMPAK FAE for assistance.

(如果模組吃錫品質考量側面爬錫，如何優化鋼網開孔設計以調整適當的錫膏量是非常重要的。尤其鋼網的厚度大約是 0.1mm 或更薄時，可考慮局部加厚鋼網的設計。請諮詢製程工程師以優化鋼網的設計,或是聯絡正基科技技術支持團隊).



7. External clock reference

External LPO signal characteristics

Parameter	Specification	Units
Nominal input frequency	32.768	kHz
Frequency accuracy	+/-30	ppm
Duty cycle	30 - 70	%
Input signal amplitude	400 to 3300	mV, p-p
Signal type	Square-wave or sine-wave	-
Input impedance	>100k	Ω
	<5	pF
Clock jitter (integrated over 300Hz – 15KHz)	<1	Hz
Output high voltage	0.7V _{io} - V _{io}	V

Input signal amplitude follow VDDIO (1.8V or 3.3V)

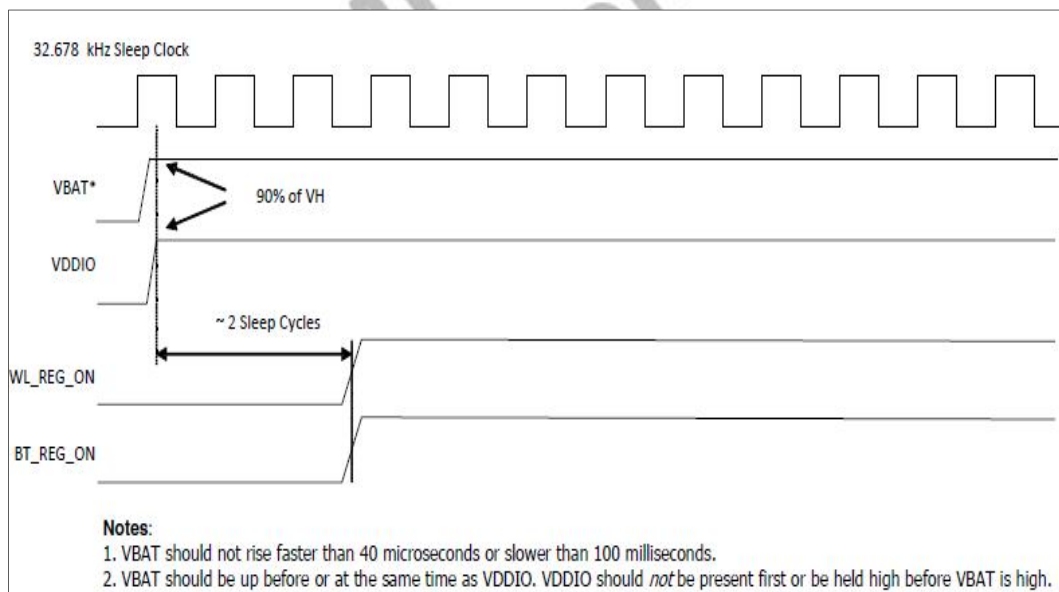
8. Host Interface Timing Diagram

8.1 Power-up Sequence Timing Diagram

The module has signals that allow the host to control power consumption by enabling or disabling the Bluetooth, WLAN and internal regulator blocks. These signals are described below.

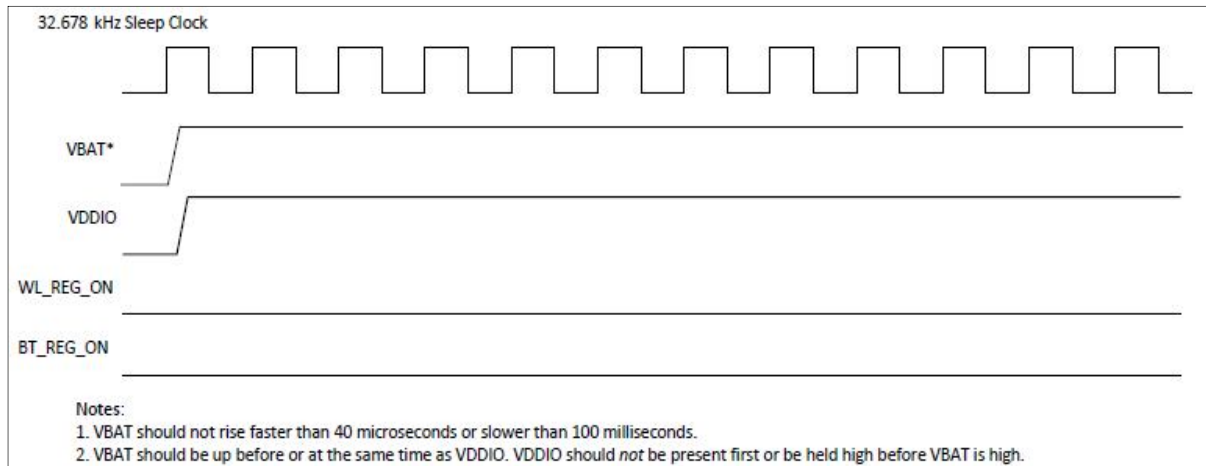
Additionally, diagrams are provided to indicate proper sequencing of the signals for various operating states. The timing value indicated are minimum required values: longer delays are also acceptable.

- **WL_REG_ON**: Used by the PMU to power up or power down the internal regulators used by the WLAN section. When this pin is high, the regulators are enabled and the WLAN section is out of reset. When this pin is low the WLAN section is in reset.
- **BT_REG_ON**: Used by the PMU to power up or power down the internal regulators used by the BT section. Low asserting reset for Bluetooth. This pin has no effect on WLAN and does not control any PMU functions. This pin must be driven high or low (not left floating).

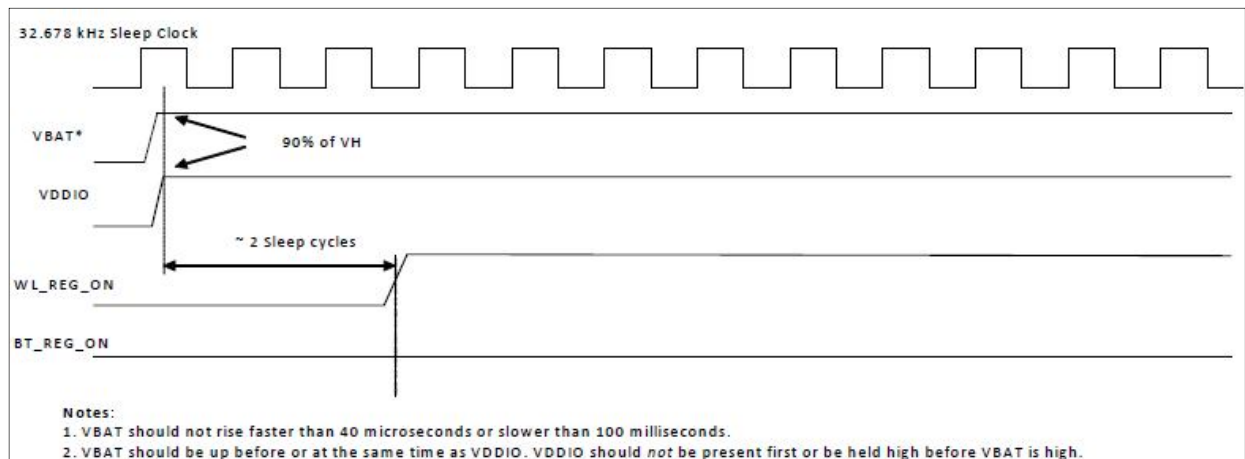


WLAN=ON, Bluetooth=ON

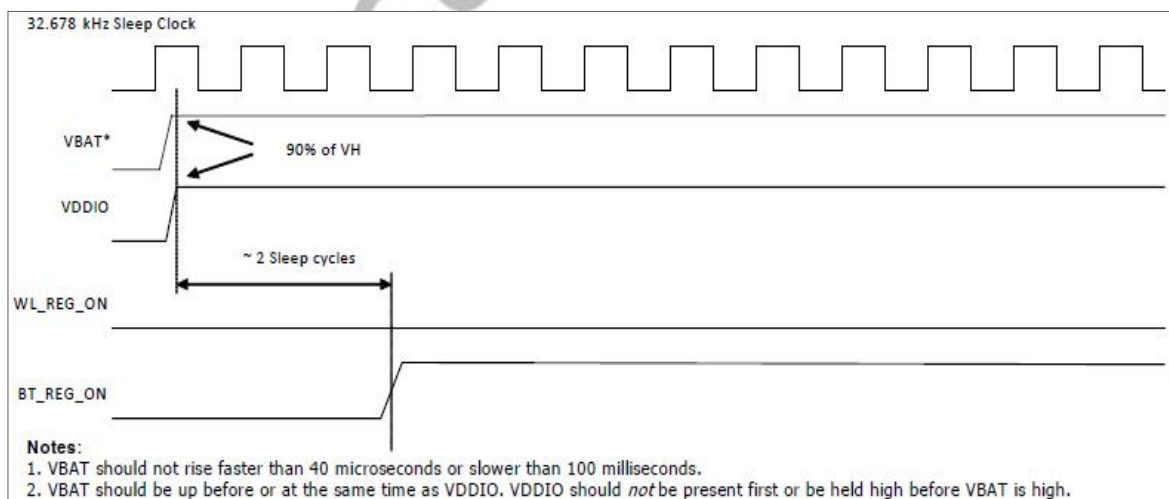




WLAN=OFF, Bluetooth=OFF



WLAN=ON, Bluetooth=OFF



WLAN=OFF, Bluetooth=ON



8.2 SDIO Interface Description

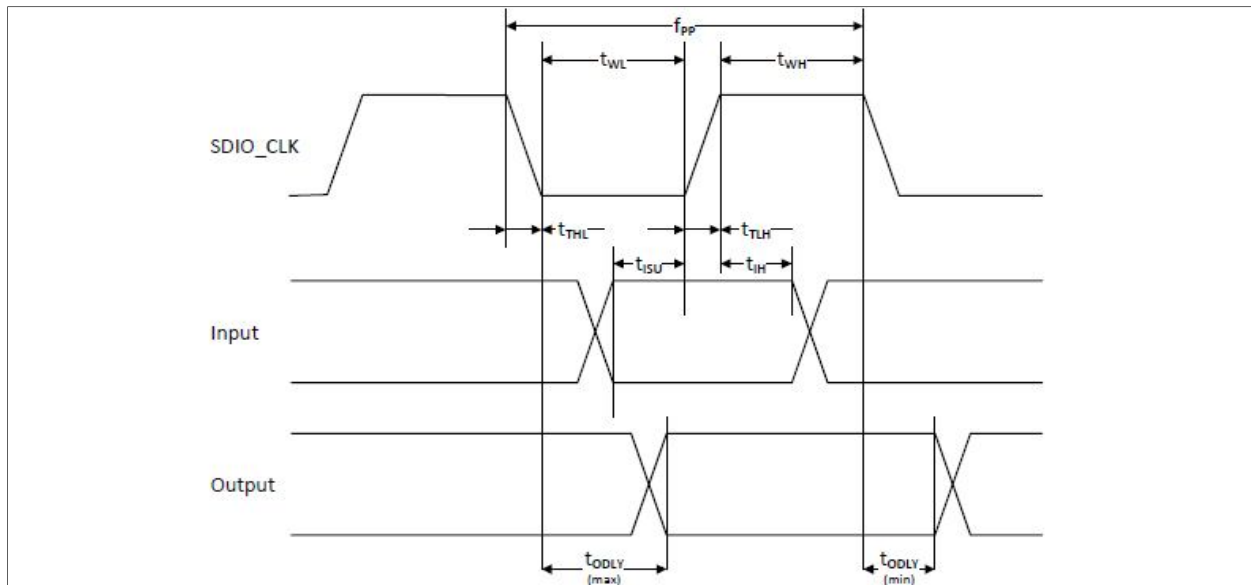
The module supports SDIO version 2.0 for both 1-bit(25 Mbps) and 4-bit modes(100 Mbps), as well as high speed 4-bit mode(50MHz clock – 200 Mbps).

- Function 0 Standard SDIO function (Max Block Size / Byte Count = 32B)
- Function 1 Backplane Function to access the internal System On Chip (SOC) address space (Max Block Size / Byte Count = 64B)
- Function 2 WLAN Function for efficient WLAN packet transfer through DMA (Max Block Size/Byte Count=512Bytes)

SDIO/gSPI Pin Description

SD 4-Bit Mode		gSPI Mode	
DATA0	Data Line 0	DO	Data Output
DATA1	Data Line 1 or Interrupt	IRQ	Interrupt
DATA2	Data Line 2 or Read Wait	NC	Not Used
DATA3	Data Line 3	CS	Card select
CLK	Clock	SCLK	Clock
CMD	Command Line	DI	Data Input

SDIO Default Mode Timing Diagram

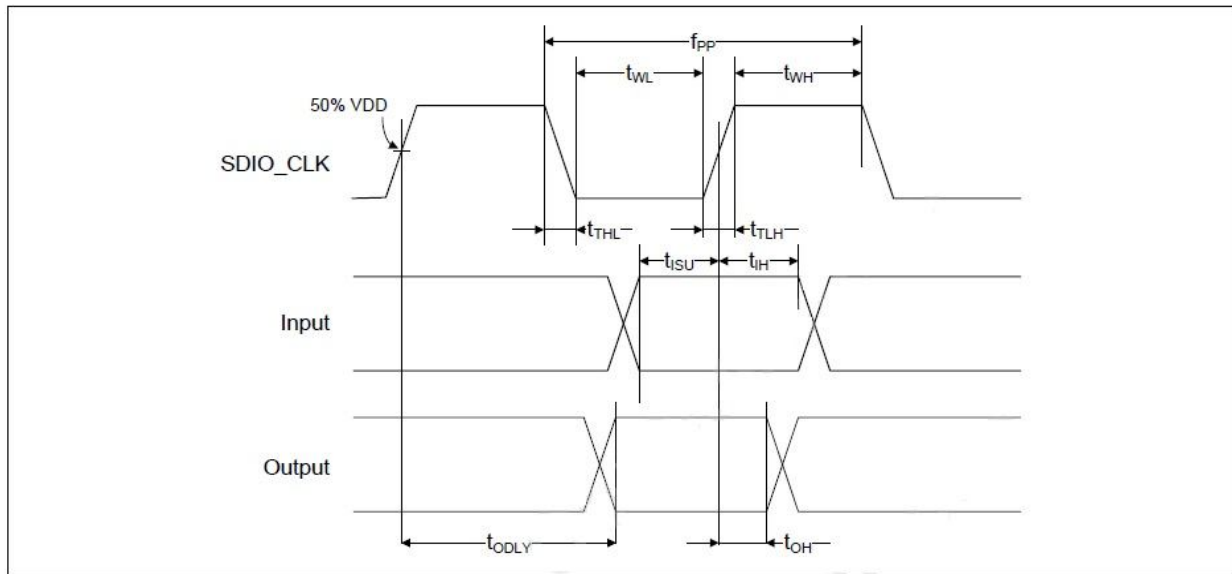


Parameter	Symbol	Minimum	Typical	Maximum	Unit
SDIO CLK (All values are referred to minimum V_{IH} and maximum V_{IL}^b)					
Frequency – Data Transfer mode	f_{PP}	0	–	25	MHz
Frequency – Identification mode	f_{OD}	0	–	400	kHz
Clock low time	t_{WL}	10	–	–	ns
Clock high time	t_{WH}	10	–	–	ns
Clock rise time	t_{TLH}	–	–	10	ns
Clock low time	t_{THL}	–	–	10	ns
Inputs: CMD, DAT (referenced to CLK)					
Input setup time	t_{ISU}	5	–	–	ns
Input hold time	t_{IH}	5	–	–	ns
Outputs: CMD, DAT (referenced to CLK)					
Output delay time – Data Transfer mode	t_{ODLY}	0	–	14	ns
Output delay time – Identification mode	t_{ODLY}	0	–	50	ns

a. Timing is based on $CL \leq 40pF$ load on CMD and Data.

b. $\min(V_{IH}) = 0.7 \times V_{DDIO}$ and $\max(V_{IL}) = 0.2 \times V_{DDIO}$.

SDIO High Speed Mode Timing Diagram



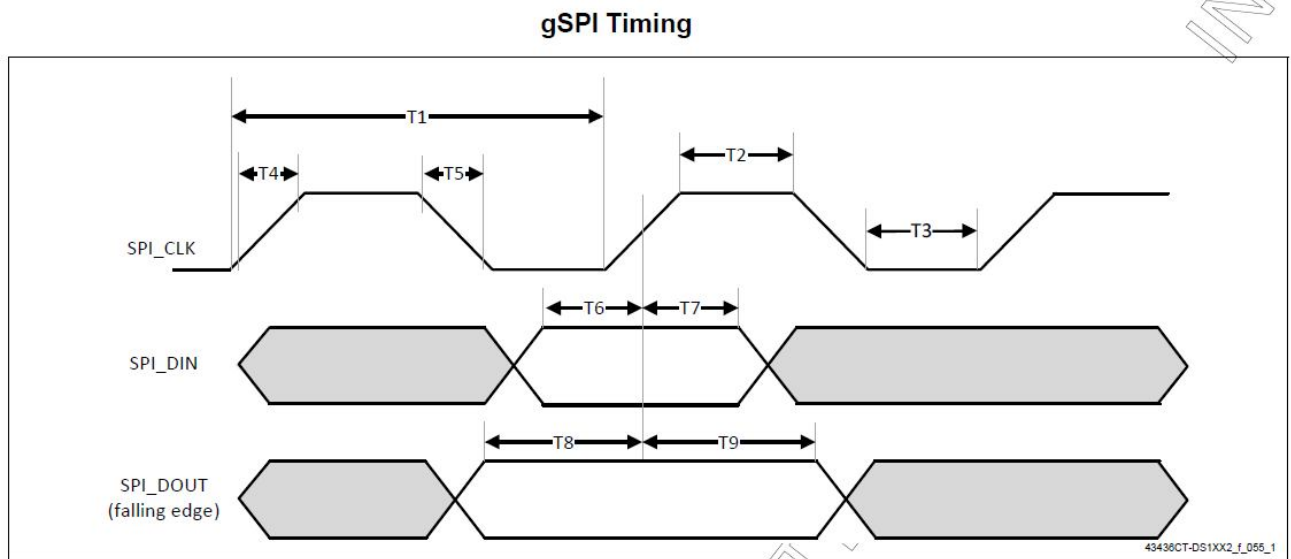
Parameter	Symbol	Minimum	Typical	Maximum	Unit
SDIO CLK (all values are referred to minimum V_{IH} and maximum V_{IL}^b)					
Frequency – Data Transfer Mode	f_{PP}	0	–	50	MHz
Frequency – Identification Mode	f_{OD}	0	–	400	kHz
Clock low time	t_{WL}	7	–	–	ns
Clock high time	t_{WH}	7	–	–	ns
Clock rise time	t_{TLH}	–	–	3	ns
Clock low time	t_{THL}	–	–	3	ns
Inputs: CMD, DAT (referenced to CLK)					
Input setup Time	t_{ISU}	6	–	–	ns
Input hold Time	t_{IH}	2	–	–	ns
Outputs: CMD, DAT (referenced to CLK)					
Output delay time – Data Transfer Mode	t_{ODLY}	–	–	14	ns
Output hold time	t_{OH}	2.5	–	–	ns
Total system capacitance (each line)	CL	–	–	40	pF

a. Timing is based on $CL \leq 40$ pF load on CMD and Data.

b. $\min(V_{IH}) = 0.7 \times V_{DDIO}$ and $\max(V_{IL}) = 0.2 \times V_{DDIO}$.



8.3 gSPI Mode Timing Diagram



gSPI Timing Parameters

Parameter	Symbol	Minimum	Maximum	Units	Note
Clock period	T1	20.8	–	ns	$F_{\max} = 50 \text{ MHz}$
Clock high/low	T2/T3	$(0.45 \times T1) - T4$	$(0.55 \times T1) - T4$	ns	–
Clock rise/fall time	T4/T5	–	2.5	ns	–
Input setup time	T6	5.0	–	ns	Setup time, SIMO valid to SPI_CLK active edge
Input hold time	T7	5.0	–	ns	Hold time, SPI_CLK active edge to SIMO invalid
Output setup time	T8	5.0	–	ns	Setup time, SOMI valid before SPI_CLK rising
Output hold time	T9	5.0	–	ns	Hold time, SPI_CLK active edge to SOMI invalid
CSX to clock ^a	–	7.86	–	ns	CSX fall to 1st rising edge
Clock to CSX	–	–	–	ns	Last falling edge to CSX high

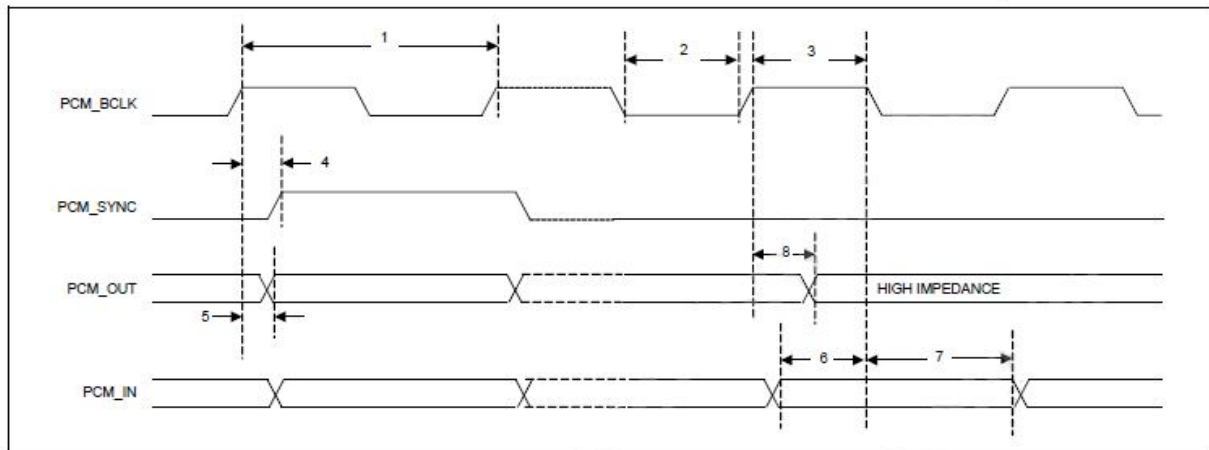
a. SPI_CSx remains active for entire duration of gSPI read/write/write_read transaction (that is, overall words for multiple word transaction).

8.5 PCM Interface Description

The PCM Interface on the BCM43456 can connect to linear PCM Codec devices in master or slave mode. In master mode, the BCM43456 generates the PCM_CLK and PCM_SYNC signals, and in slave mode, these signals are provided by another master on the PCM interface and are inputs to the BCM43456. The configuration of the PCM interface may be adjusted by the host through the use of vendor-specific HCI commands.

Short Frame Sync, Master Modem

PCM Timing Diagram (Short Frame Sync, Master Mode)



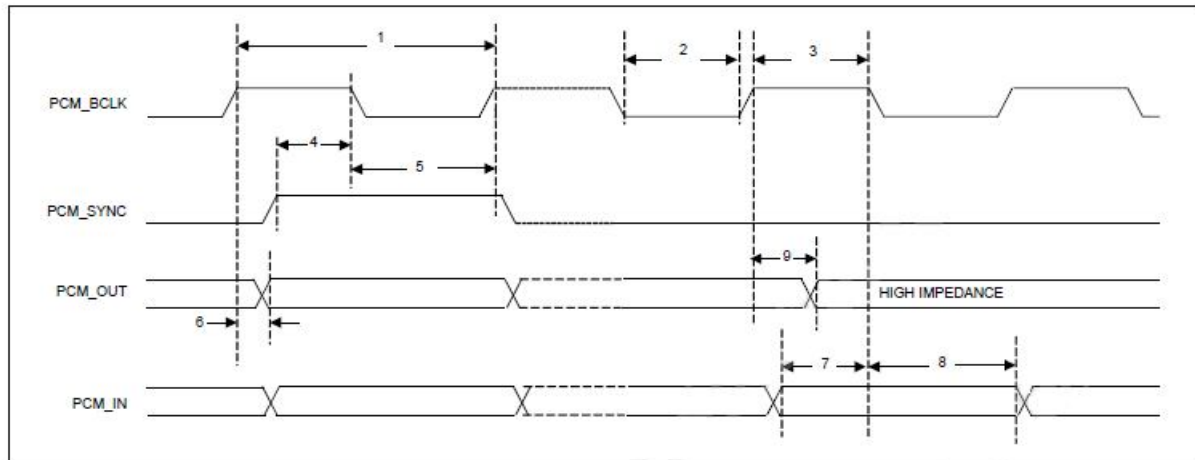
PCM Interface Timing Specifications (Short Frame Sync, Master Mode)

Reference	Characteristics	Minimum	Typical	Maximum	Unit
1	PCM bit clock frequency		–	12	MHz
2	PCM bit clock low	41	–	–	ns
3	PCM bit clock high	41	–	–	ns
4	PCM_SYNC delay	0	–	25	ns
5	PCM_OUT delay	0	–	25	ns
6	PCM_IN setup	8	–	–	ns
7	PCM_IN hold	8	–	–	ns
8	Delay from rising edge of PCM_BCLK during last bit period to PCM_OUT becoming high impedance	0	–	25	ns



Short Frame Sync, Slave Mode

PCM Timing Diagram (Short Frame Sync, Slave Mode)



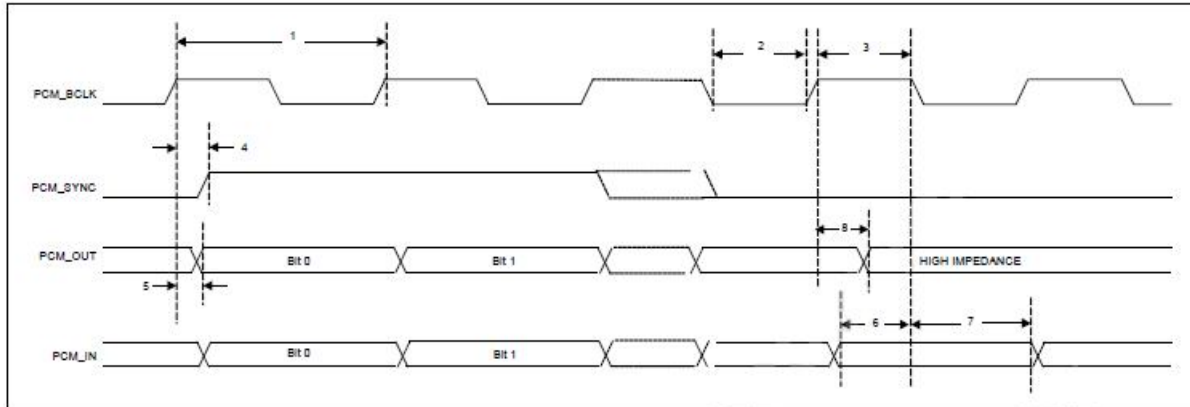
PCM Interface Timing Specifications (Short Frame Sync, Slave Mode)

Reference	Characteristics	Minimum	Typical	Maximum	Unit
1	PCM bit clock frequency	—	—	12	MHz
2	PCM bit clock low	41	—	—	ns
3	PCM bit clock high	41	—	—	ns
4	PCM_SYNC setup	8	—	—	ns
5	PCM_SYNC hold	8	—	—	ns
6	PCM_OUT delay	0	—	25	ns
7	PCM_IN setup	8	—	—	ns
8	PCM_IN hold	8	—	—	ns
9	Delay from rising edge of PCM_BCLK during last bit period to PCM_OUT becoming high impedance	0	—	25	ns



Long Frame Sync, Master Mode

PCM Timing Diagram (Long Frame Sync, Master Mode)



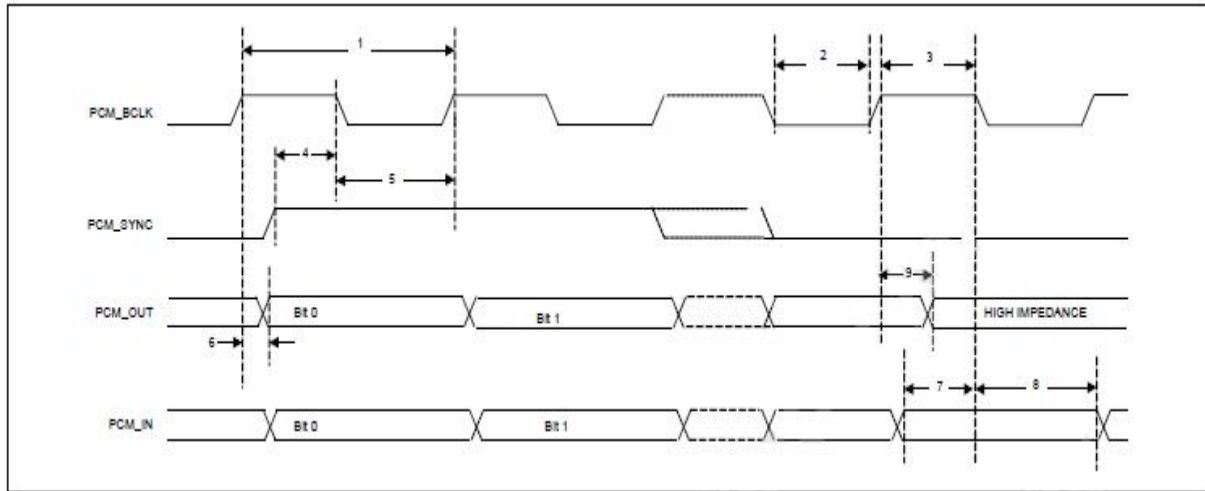
PCM Interface Timing Specifications (Long Frame Sync, Master Mode)

Reference	Characteristics	Minimum	Typical	Maximum	Unit
1	PCM bit clock frequency	–	–	12	MHz
2	PCM bit clock low	41	–	–	ns
3	PCM bit clock high	41	–	–	ns
4	PCM_SYNC delay	0	–	25	ns
5	PCM_OUT delay	0	–	25	ns
6	PCM_IN setup	8	–	–	ns
7	PCM_IN hold	8	–	–	ns
8	Delay from rising edge of PCM_BCLK during last bit period to PCM_OUT becoming high impedance	0	–	25	ns



Long Frame Sync, Slave Mode

PCM Timing Diagram (Long Frame Sync, Slave Mode)



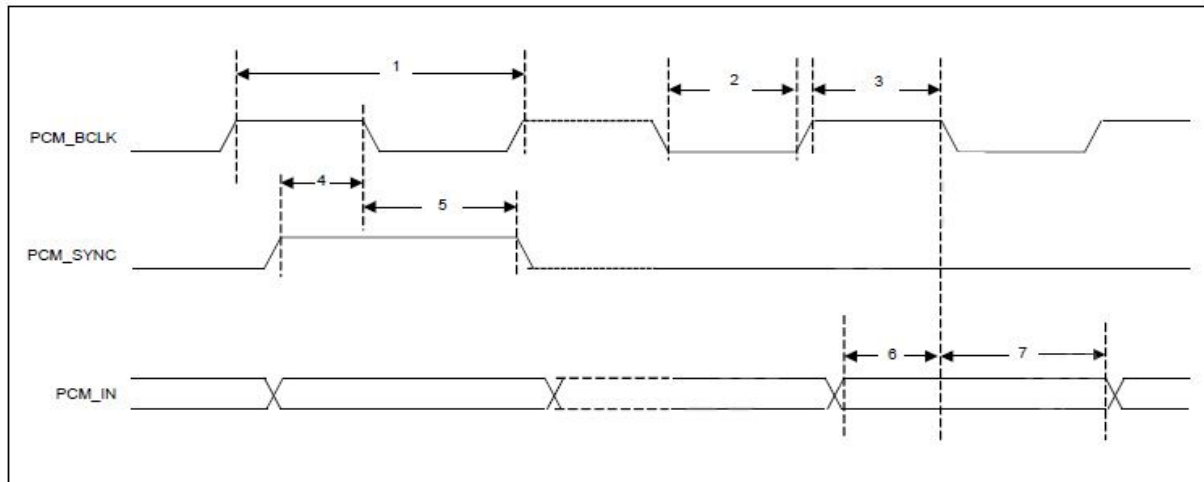
PCM Interface Timing Specifications (Long Frame Sync, Slave Mode)

Reference	Characteristics	Minimum	Typical	Maximum	Unit
1	PCM bit clock frequency	–	–	12	MHz
2	PCM bit clock low	41	–	–	ns
3	PCM bit clock high	41	–	–	ns
4	PCM_SYNC setup	8	–	–	ns
5	PCM_SYNC hold	8	–	–	ns
6	PCM_OUT delay	0	–	25	ns
7	PCM_IN setup	8	–	–	ns
8	PCM_IN hold	8	–	–	ns
9	Delay from rising edge of PCM_BCLK during last bit period to PCM_OUT becoming high impedance	0	–	25	ns



Short Frame Sync, Burst Mode

PCM Burst Mode Timing (Receive Only, Short Frame Sync)



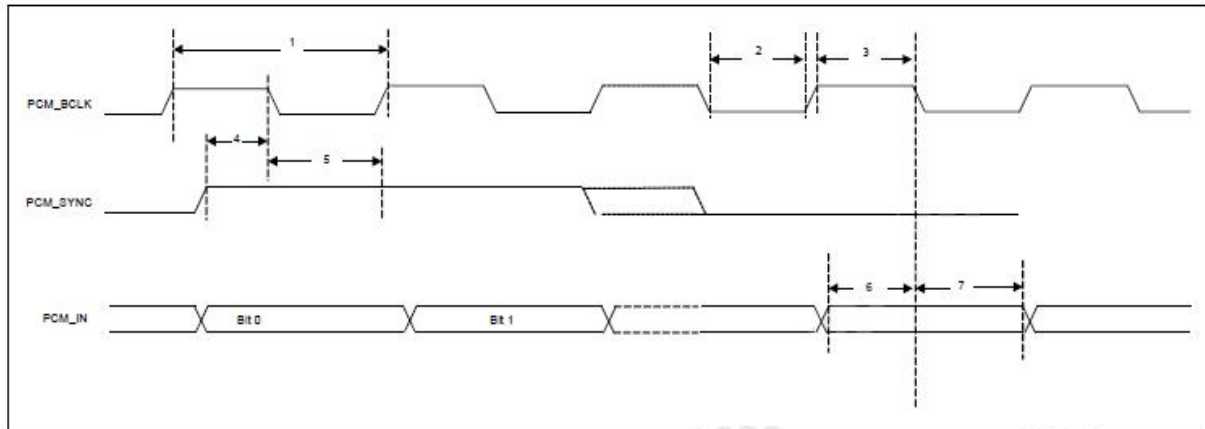
PCM Burst Mode (Receive Only, Short Frame Sync)

Reference	Characteristics	Minimum	Typical	Maximum	Unit
1	PCM bit clock frequency	–	–	24	MHz
2	PCM bit clock low	20.8	–	–	ns
3	PCM bit clock high	20.8	–	–	ns
4	PCM_SYNC setup	8	–	–	ns
5	PCM_SYNC hold	8	–	–	ns
6	PCM_IN setup	8	–	–	ns
7	PCM_IN hold	8	–	–	ns



Long Frame Sync, Burst Mode

PCM Burst Mode Timing (Receive Only, Long Frame Sync)



PCM Burst Mode (Receive Only, Long Frame Sync)

Reference	Characteristics	Minimum	Typical	Maximum	Unit
1	PCM bit clock frequency	–	–	24	MHz
2	PCM bit clock low	20.8	–	–	ns
3	PCM bit clock high	20.8	–	–	ns
4	PCM_SYNC setup	8	–	–	ns
5	PCM_SYNC hold	8	–	–	ns
6	PCM_IN setup	8	–	–	ns
7	PCM_IN hold	8	–	–	ns



8.6 UART Interface Description

The UART is a standard 4-wire interface (RX, TX, RTS, and CTS) with adjustable baud rates from 9600 bps to 4.0 Mbps. The interface features an automatic baud rate detection capability that returns a baud rate selection. Alternatively, the baud rate may be selected through a vendor-specific UART HCI command.

UART has a 1040-byte receive FIFO and a 1040-byte transmit FIFO to support EDR. Access to the FIFOs is conducted through the AHB interface through either DMA or the CPU. The UART supports the Bluetooth 4.2 UART HCI specification: H4, a custom Extended H4, and H5. The default baud rate is 115.2 Kbaud.

The UART supports the 3-wire H5 UART transport, as described in the Bluetooth specification (Three-wire UART Transport Layer). Compared to H4, the H5 UART transport reduces the number of signal lines required by eliminating the CTS and RTS signals.

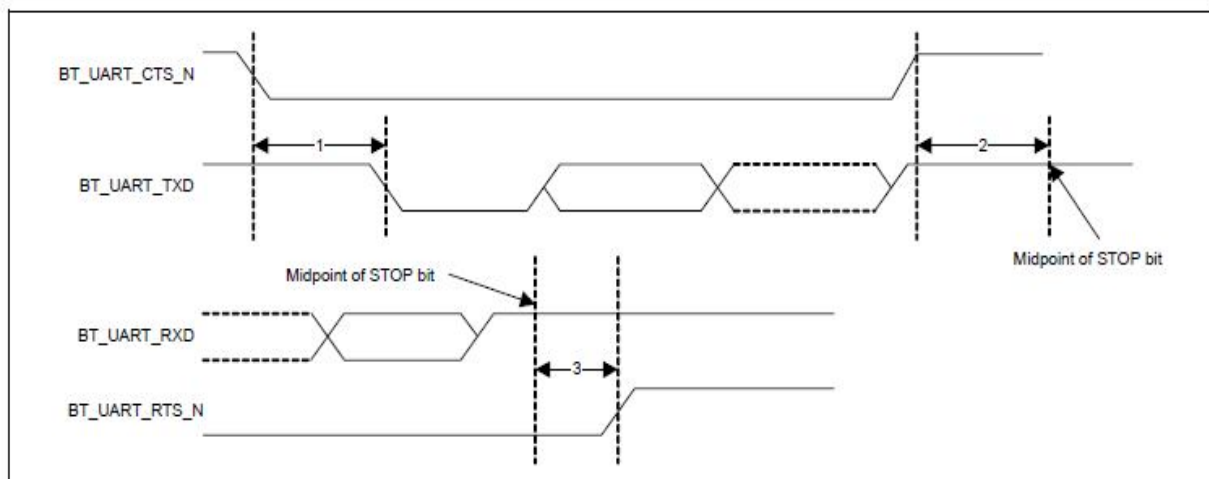
The BCM43456 UART can perform XON/XOFF flow control and includes hardware support for the Serial Line Input Protocol (SLIP). It can also perform wake-on activity. For example, activity on the RX or CTS inputs can wake the chip from a sleep state.

Normally, the UART baud rate is set by a configuration record downloaded after device reset, or by automatic baud rate detection, and the host does not need to adjust the baud rate. Support for changing the baud rate during normal HCI UART operation is included through a vendor-specific command that allows the host to adjust the contents of the baud rate registers. The BCM43456 UARTs operate correctly with the host UART as long as the combined baud rate error of the two devices is within $\pm 2\%$.

Example of Common Baud Rates

Desired Rate	Actual Rate	Error (%)
4000000	4000000	0.00
3692000	3692308	0.01
3000000	3000000	0.00
2000000	2000000	0.00
1500000	1500000	0.00
1444444	1454544	0.70
921600	923077	0.16
460800	461538	0.16
230400	230796	0.17
115200	115385	0.16
57600	57692	0.16
38400	38400	0.00
28800	28846	0.16
19200	19200	0.00
14400	14423	0.16
9600	9600	0.00

UART Timing

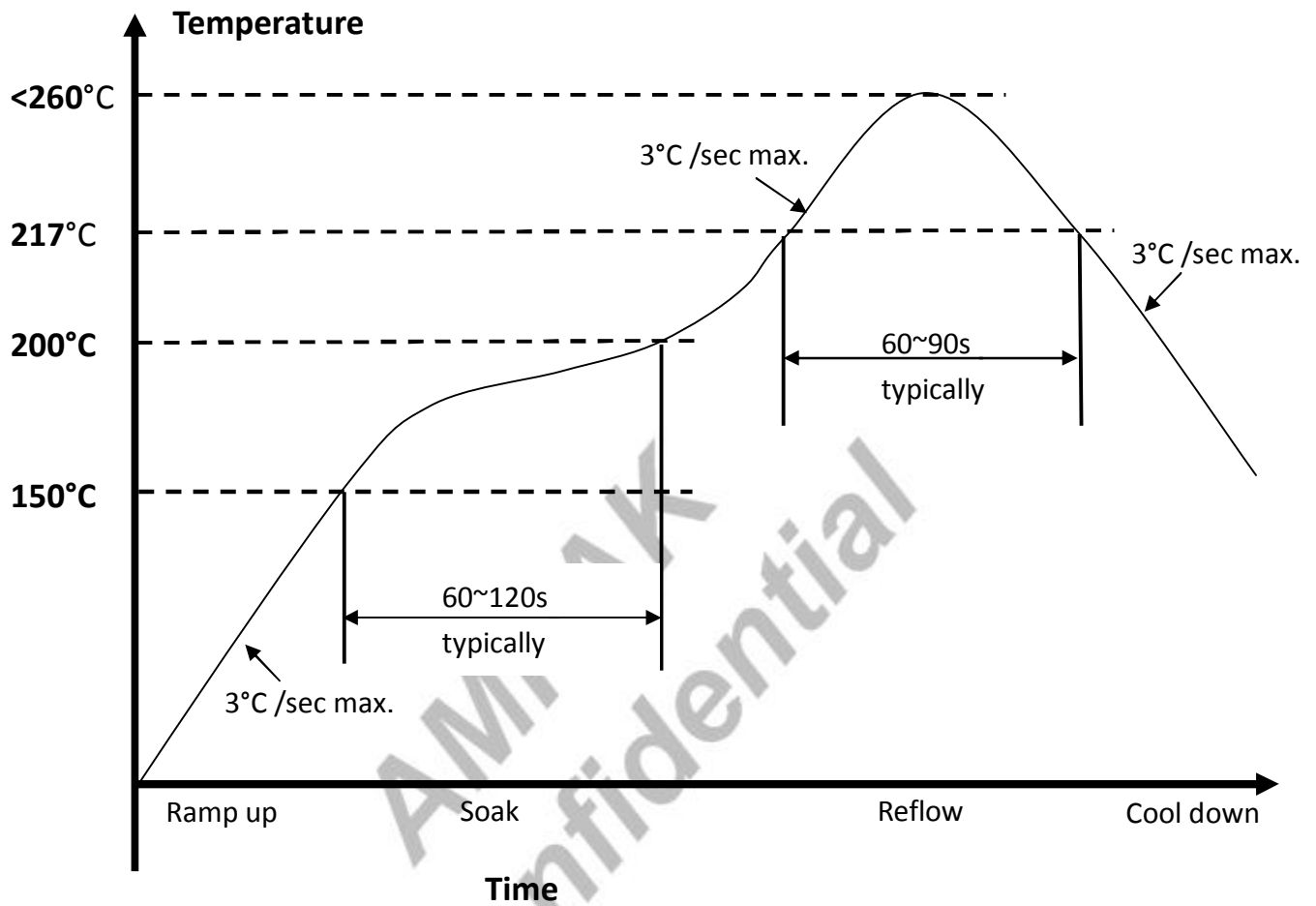


UART Timing Specifications

Ref	Characteristics	Min.	Typ.	Max.	Unit
1	Delay time, BT_UART_CTS_N low to BT_UART_TXD valid	–	–	1.5	Bit periods
2	Setup time, BT_UART_CTS_N high before midpoint of stop bit	–	–	0.5	Bit periods
3	Delay time, midpoint of stop bit to BT_UART_RTS_N high	–	–	0.5	Bit periods



9. Recommended Reflow Profile



1. Referred to IPC/JEDEC standard
2. Peak Temperature : <260°C
3. Cycle of Reflow : 2 times max.
4. Adding Nitrogen (N₂) to implement 2000ppm or less of oxygen concentration during reflow process is recommended.
5. If the shelf time is exceeded, be sure baking step to remove the moisture from the component.

10. Package Information

10.1 Label

Label A → Anti-static and humidity notice



Label B → MSL caution / Storage Condition

Caution		LEVEL
This bag contains MOISTURE-SENSITIVE DEVICES		 If blank, see adjacent bar code label
1. Calculated shelf life in sealed bag: 12 months at <40°C and <90% relative humidity (RH) 2. Peak package body temperature: _____ °C If blank, see adjacent bar code label 3. After bag is opened, devices that will be subjected to reflow solder or other high temperature process must be a) Mounted within: _____ hours of factory conditions If blank, see adjacent bar code label ≤30°C/60% RH, or b) Stored per J-STD-033 4. Devices require bake, before mounting, if: a) Humidity Indicator Card reads >10% for level 2a - 5a devices or >60% for level 2 devices when read at 23 ± 5°C b) 3a or 3b are not met 5. If baking is required, refer to IPC/JEDEC J-STD-033 for bake procedure Bag Seal Date: _____ If blank, see adjacent bar code label Note: Level and body temperature defined by IPC/JEDEC J-STD-020		

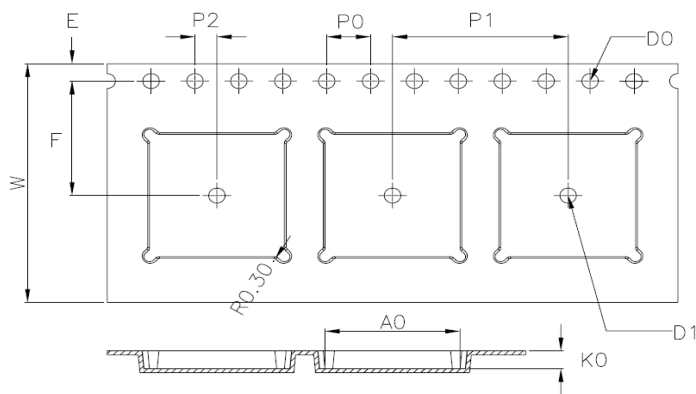
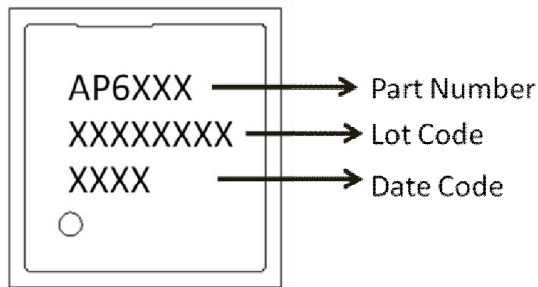
Label C → Inner box label .

PO:	
AMK DEVICE:	
PKG S/N:	 9PKGYMDDNNNNN
Model Name:	 APXXXXXXX (R3HF)
P/N:	 99X-XXX-XXXXR
Quantity:	 QQQQ
Date Code:	 YYWW
Lot Code:	 XXXXXXXX

Label D → Carton box label .

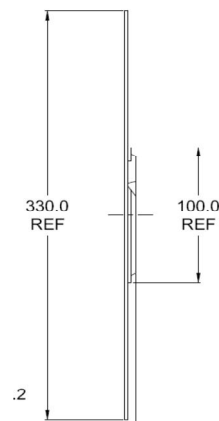
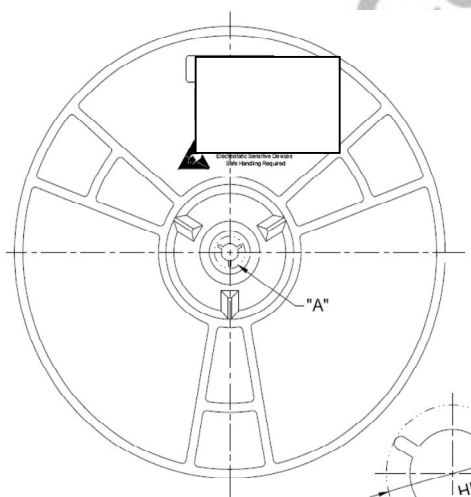
AMPAK Technology Inc.	
PO:	
AMK DEVICE:	
Model Name:	 APXXXXXXX (R3HF)
Part No.:	 99X-XXX-XXXXR
Quantity:	 QQQQ
Lot D/C:	 XXXXXXXX YYWW QQQQ
Manufacture:	 YYYY/MM/DD

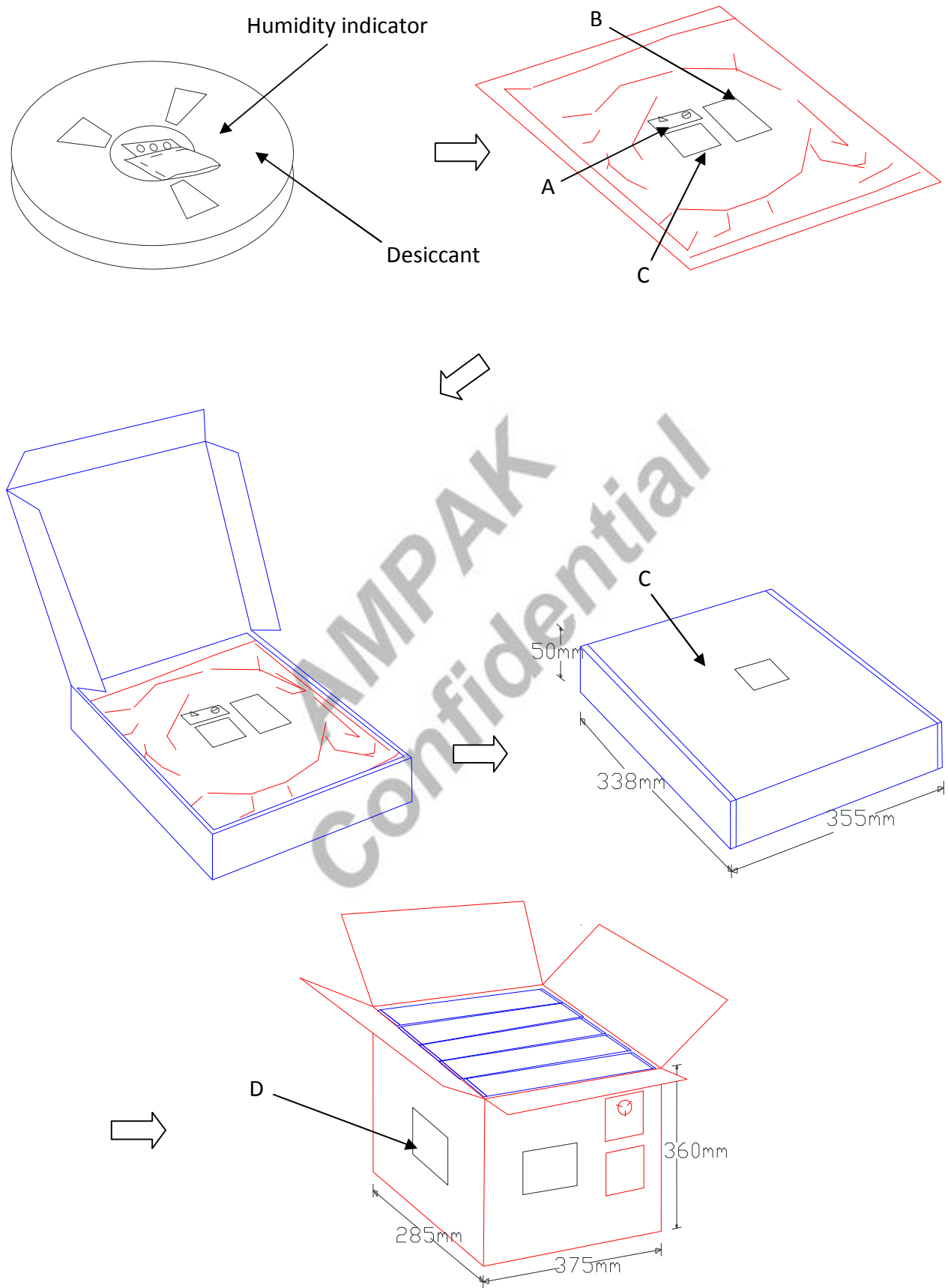
10.2 Dimension



W	24.00±0.30
A0	12.30±0.10
B0	12.30±0.10
K0	1.80±0.10
E	1.75±0.10
F	11.50±0.10
P0	4.00±0.10
P1	16.00±0.10
P2	2.00±0.10
D0	1.50 $\begin{smallmatrix} +0.10 \\ -0.00 \end{smallmatrix}$
D1	Ø1.50MIN

1. 10 sprocket hole pitch cumulative tolerance ± 0.20 .
2. Carrier camber is within 1 mm in 250 mm.
3. Material: Black Conductive Polystyrene Alloy.
4. All dimensions meet EIA-481-D requirements.
5. Thickness: 0.30 ± 0.05 mm.
6. Component load per 13" reel : 1500 PCS





10.3 MSL Level / Storage Condition

	Caution This bag contains MOISTURE-SENSITIVE DEVICES	LEVEL 4 If blank, see adjacent bar code label
1. Calculated shelf life in sealed bag: 12 months at $<40^{\circ}\text{C}$ and $<90\%$ relative humidity (RH) 2. Peak package body temperature: <u>250</u> $^{\circ}\text{C}$ If blank, see adjacent bar code label 3. After bag is opened, devices that will be subjected to reflow solder or other high temperature process must be a) Mounted within: <u>72</u> hours of factory conditions If blank, see adjacent bar code label $\leq 30^{\circ}\text{C}/60\% \text{ RH}$, or b) Stored per J-STD-033 4. Devices require bake, before mounting, if: a) Humidity Indicator Card reads $>10\%$ for level 2a-5a devices or $>60\%$ for level 2 devices when read at $23 \pm 5^{\circ}\text{C}$ b) 3a or 3b are not met. 5. If baking is required, refer to IPC/JEDEC J-STD-033 for bake procedure.		
Bag Seal Date: _____ If blank, see adjacent bar code label		
Note: Level and body temperature defined by IPC/JEDEC J-STD-020		